

## Does antibacterial treatment for urinary tract infection contribute to the risk of breast cancer?

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Low lignan status has been reported to be related to an elevated risk of breast cancer. Since lignan status is reduced by antibacterial medications, it is plausible to hypothesize that repeated use of antibiotics may also be a risk factor for breast cancer. History of treatment for urinary tract infection was studied for its prediction of breast cancer among 9461 Finnish women 19–89 years of age and initially cancer-free. During a follow-up in 1973–1991, a total of 157 breast cancer cases were diagnosed. Women reporting previous or present medication for urinary tract infection at baseline showed an elevated breast cancer risk in comparison with other women. The age-adjusted relative risk was 1.34 (95% confidence interval (CI) = 0.98–1.83). The association was concentrated to women under 50 years of age. The relative risk for these women was 1.74 (95% CI 1.13–2.68), whereas it was 0.97 (95% CI 0.59–1.58) for older women. The relative risk in the younger age-group was 1.47 (95% CI 0.73–2.97) during the first 10 years of follow-up, and 1.93 (95% CI 1.11–3.37) for follow-up times longer than 10 years. These data suggest that premenopausal women using long-term medication for urinary tract infections show a possible elevated risk of future breast cancer. The results are, however, still inconclusive and the hypothesis needs to be tested by other studies. © 2000 Cancer Research Campaign